

## MINI PROJECT

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# LAMP Stack Setup and Configuration for Dynamic Web Applications

## Overview

A LAMP stack is a group of four open-source software components that form the foundation of building and running high-performance dynamic websites and web applications. It's an acronym for Linux, Apache, MySQL/MariaDB and PHP.

The LAMP stack is simple, stable, and powerful, and its first layer is composed of the popular Linux Operating System with flexible configuration options and tight security. Apache is the second layer in the pile and offers you compiled module support for major scripting languages such as PHP, Perl, Python, Node.js, etc. This HTTP web server delivers web content through the internet. When it comes to the database layer, you've two options when deploying a LAMP stack. You can either use MySQL or MariaDB as an Relational Database Management System (RDMS) for your web applications. The last package in the list is PHP. This is a scripting language that you can easily embed in HTML documents to create static or dynamic websites.

The majority of modern Content Management Systems (CMS) such as WordPress, Joomla, Drupal, and Magento require a LAMP stack to work. Therefore, in this guide, you'll learn how to configure the right environment for them. The same setup will also work for any custom web application that you may intend to build with the LAMP stack technology.

## Components of the LAMP Stack

1. **Linux :** Linux is the foundation of the LAMP stack. It is a secure and reliable operating system widely used for servers and cloud environments.
2. **Apache:** Apache is a widely used web server that delivers web content to users over the internet. It handles HTTP requests and serves websites efficiently.
3. **MySQL/MariaDB:** The database layer of the LAMP stack stores and manages application data.
  - a. MySQL is one of the most popular relational database management systems (RDBMS).
  - b. MariaDB is a community-developed fork of MySQL that offers improved performance and compatibility.
4. **PHP:** PHP is a server-side scripting language used to create dynamic web applications. PHP code can be embedded directly into HTML files.

## How the LAMP Stack Works

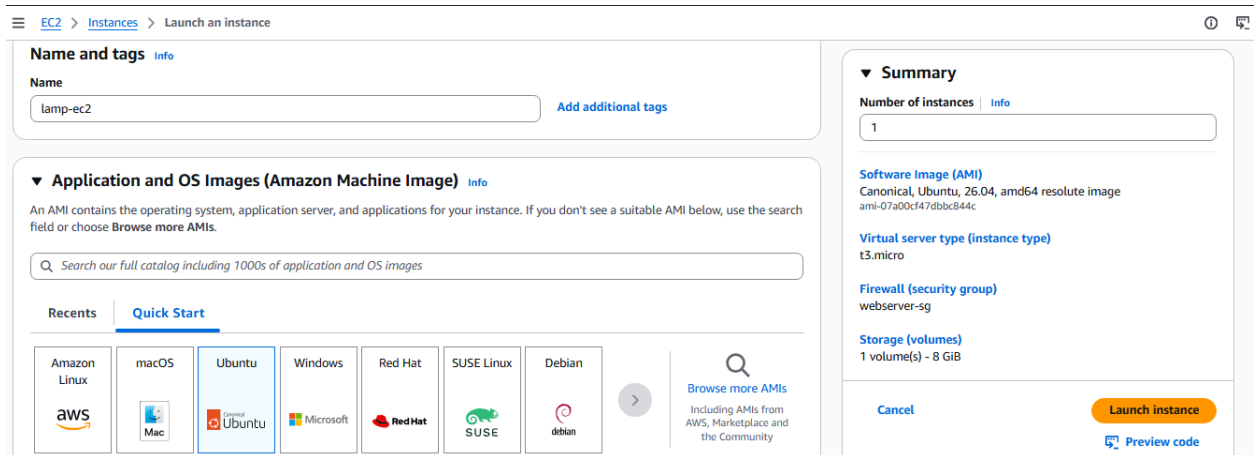
The components of the LAMP stack work together in the following way:

1. A user sends a request through a web browser.
2. Apache receives the request.
3. PHP processes the application logic.
4. MySQL/MariaDB stores or retrieves data.
5. Apache sends the final web page back to the user.

This process allows websites to generate dynamic content based on user interactions.

## Installation and Configuration

1. Launch an ec2 instance with ubuntu 20.04 or above



Here I have taken Ubuntu, 26.04

2. Install Apache Webserver

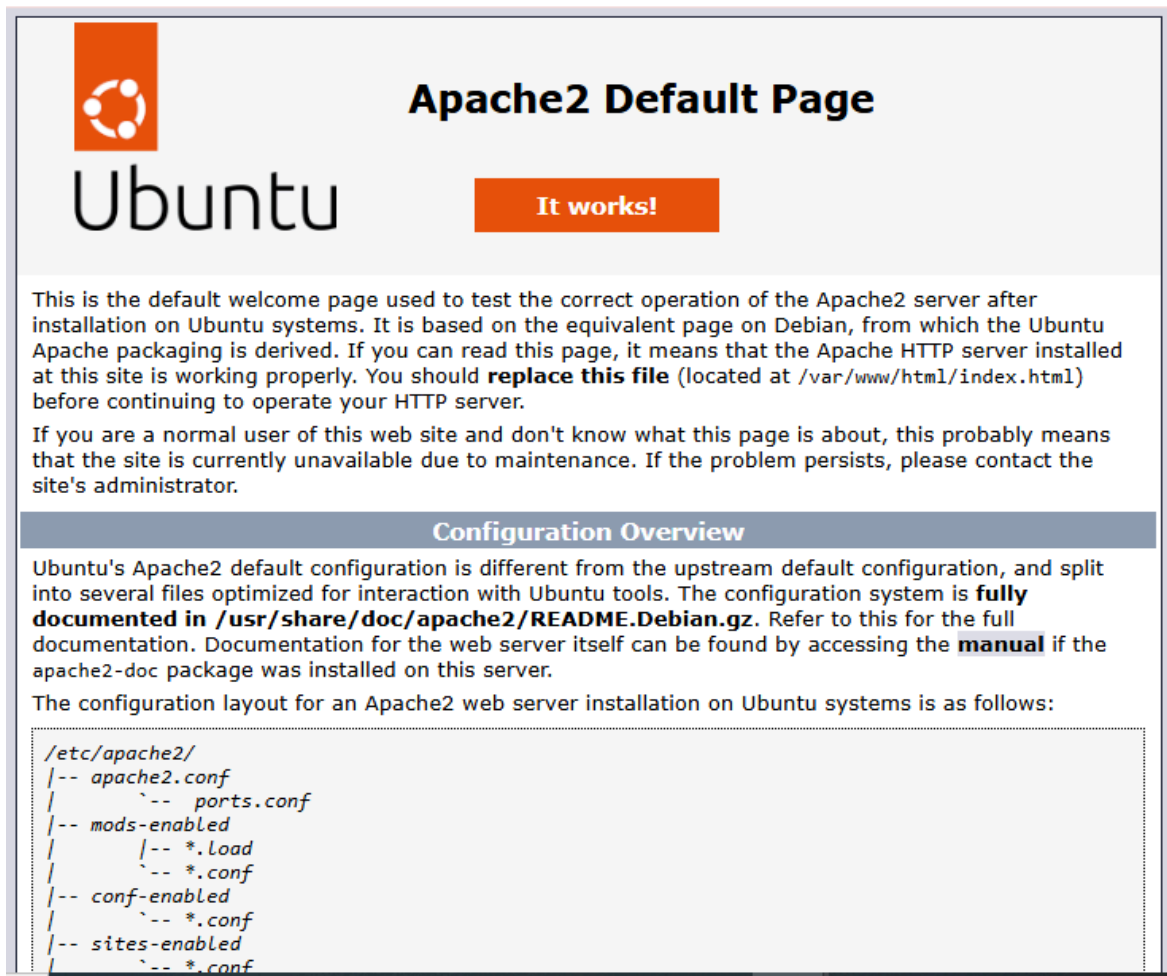
SSH to your Ubuntu server as a non-root user, then update the package information index and upgrade your packages.

```
$ sudo apt update && sudo apt -y upgrade
```

Next, run the command below to install the Apache Web server.

```
$ sudo apt install -y apache2
```

Visit the URL `http://<public IP address of your server or domain name>` on a web browser you should see the default Apache web page as shown below:



After setting up the web server, you'll install a database in the next step.

### 3. Install a Database Server

- You can either install MySQL or MariaDB server when deploying a LAMP stack. Here I have used MariaDB, run the command below.

```
$ sudo apt install -y mariadb-server mariadb-client
```

- Irrespective of the database that you've chosen, run the command below to secure it.

```
$ sudo mariadb-secure-installation
```

- c. Enter the below choices and press Enter in each prompt to proceed.

```
Enter current password for root (enter for none):  
:key enter:
```

```
Set root password? [Y/n]: Y
```

```
New password: EXAMPLE PASSWORD
```

```
Re-enter new password: EXAMPLE PASSWORD
```

```
Remove anonymous users? [Y/n] Y
```

```
Disallow root login remotely? [Y/n] Y
```

```
Remove test database and access to it? [Y/n] Y
```

```
Reload privilege tables now? [Y/n] Y
```

- d. Once you've done the installation, login to the MariaDB as root as below:

```
$ sudo mariadb -u root -p
```

- e. Enter your database server root password and press Enter to proceed. Then, type the command below to create your first test\_database database.

```
MariaDB [(none)]> CREATE database test_database;
```

Output is shown below:

```

type 'help;' or '\h' for help. Type '\c' to clear the
MariaDB [(none)]> CREATE database test_database;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql          |
| performance_schema |
| sys            |
| test_database   |
+-----+
5 rows in set (0.001 sec)

MariaDB [(none)]>

```

f. `i.L01d746f48bf5c1a1d (/lammp-ubuntu-ec2)`

- g. Create a test\_user and assign full privileges to the database you've just created. Replace EXAMPLE\_PASSWORD with a strong value.

```

MariaDB [(none)]> CREATE USER 'test_user'@'localhost' IDENTIFIED BY
'EXAMPLE_PASSWORD';

```

```

GRANT ALL PRIVILEGES ON test_database.* TO 'test_user'@'localhost';

```

```

FLUSH PRIVILEGES;

```

```

EXIT;

```

- h. Your database server is now ready

#### 4. Install PHP

- a. In this step, you'll install the PHP package. Run the command below.

```

$ sudo apt install -y php

```

```

$ sudo apt install -y

```

```

php-{common,mysql,xm1,xm1rpc,curl,gd,imagick,cli,dev,imap,mbstring,opcache,so
ap,zip,intl}

```

- b. Restart the Apache webserver to load PHP.

```

$ sudo systemctl restart apache2

```

- c. To test PHP, create an info.php file in the root directory of your web server.

```
$ sudo nano /var/www/html/info.php
```

- d. Then, enter the information below into the file.

```
<?php
```

```
phpinfo();
```

- e. Then, in a web browser, visit the URL <http://<public IP address>/info.php> You should get a detailed PHP page as shown below.

| PHP Version 8.5.4                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| System                                                                              | Linux ip-10-0-0-234 7.0.0-1004-aws #4-Ubuntu SMP PREEMPT Mon Apr 13 13:14:24 UTC 2026 x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Build Date                                                                          | Apr 1 2026 09:36:11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Build System                                                                        | Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Build Provider                                                                      | Ubuntu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Server API                                                                          | Apache 2 Handler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Virtual Directory Support                                                           | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Configuration File (php.ini) Path                                                   | /etc/php/8.5/apache2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Loaded Configuration File                                                           | /etc/php/8.5/apache2/php.ini                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Scan this dir for additional .ini files                                             | /etc/php/8.5/apache2/conf.d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Additional .ini files parsed                                                        | /etc/php/8.5/apache2/conf.d/10-pdo.ini, /etc/php/8.5/apache2/conf.d/20-calendar.ini, /etc/php/8.5/apache2/conf.d/20-ctype.ini, /etc/php/8.5/apache2/conf.d/20-enf.ini, /etc/php/8.5/apache2/conf.d/20-ffi.ini, /etc/php/8.5/apache2/conf.d/20-fileinfo.ini, /etc/php/8.5/apache2/conf.d/20-ftp.ini, /etc/php/8.5/apache2/conf.d/20-gettext.ini, /etc/php/8.5/apache2/conf.d/20-iconv.ini, /etc/php/8.5/apache2/conf.d/20-igmp.ini, /etc/php/8.5/apache2/conf.d/20-sockets.ini, /etc/php/8.5/apache2/conf.d/20-readline.ini, /etc/php/8.5/apache2/conf.d/20-shmop.ini, /etc/php/8.5/apache2/conf.d/20-sockets.ini, /etc/php/8.5/apache2/conf.d/20-sysmsg.ini, /etc/php/8.5/apache2/conf.d/20-syssem.ini, /etc/php/8.5/apache2/conf.d/20-sysvshm.ini, /etc/php/8.5/apache2/conf.d/20-tokenizer.ini |
| PHP API                                                                             | 20250925                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| PHP Extension                                                                       | 20250925                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Zend Extension                                                                      | 420250925                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Zend Extension Build                                                                | API420250925.NTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PHP Extension Build                                                                 | API420250925.NTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PHP Integer Size                                                                    | 64 bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

5. Next, test PHP connectivity with the database that you created earlier.

- a. Open below file

```
$ sudo nano /var/www/html/database_test.php
```

- b. Then, enter the information below into the file.

```
<?php
```

```
$conn = new mysqli('localhost', 'test_user', 'EXAMPLE_PASSWORD', 'test_database');
```

```
if ($conn->connect_error) {
```

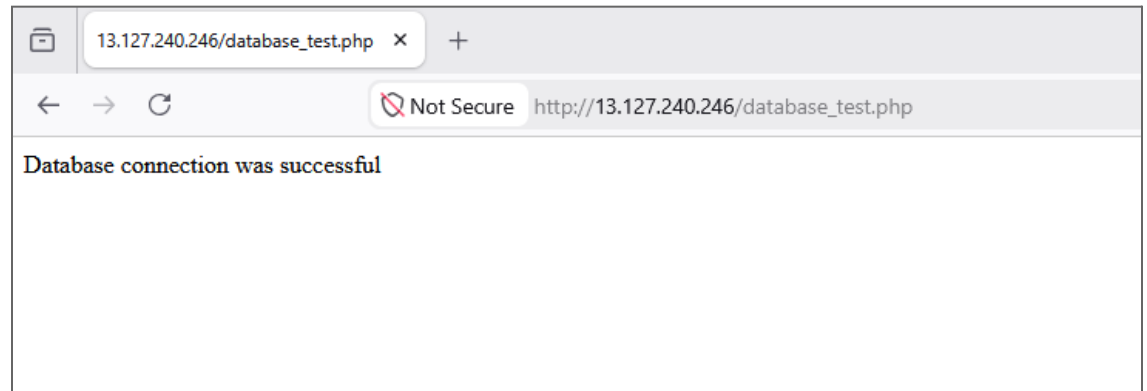
```
die("Database connection failed: " . $conn->connect_error);
```

```
}
```

- c. Save and close the file. Then, visit the address below:

`http://<public ip of your instance>/database_test.php`

- d. You should get the output below that shows that the script has successfully connected to the database.



## Conclusion

Your LAMP stack is now ready on Ubuntu, and you can now go ahead and build a dynamic website or cloud-based web application.

The LAMP stack remains one of the most powerful and widely used web development platforms in the world. By combining Linux, Apache, MySQL/MariaDB, and PHP, developers can create secure, scalable, and dynamic web applications efficiently.